

A Minor Project Synopsis

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AI-Based Routing Algorithm for Network-on-Chip (NoC)

1. INTRODUCTION

In modern computing systems, the number of cores inside a processor is increasing rapidly. To enable communication between these cores, an efficient communication system is required.

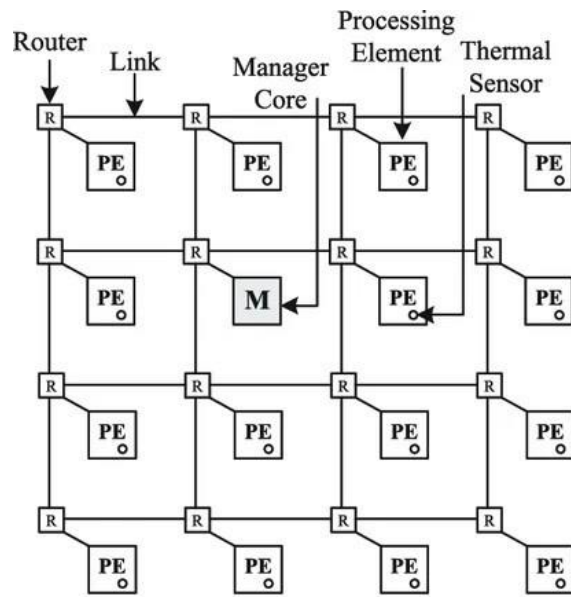
Network-on-Chip (NoC) is an advanced communication architecture that replaces traditional bus-based communication systems. In a NoC system, communication happens through a network of routers and links. Each core is connected to a router, and data is transmitted in the form of packets. This approach improves scalability and allows multiple data transfers to occur simultaneously.

However, as the number of cores increases, network congestion and delays become major issues.

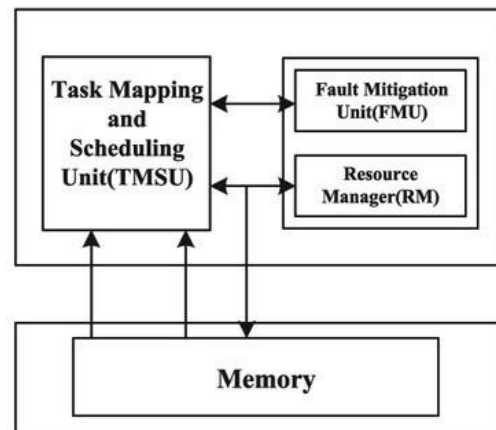
Traditional routing algorithms such as XY routing follow fixed paths and do not consider current network conditions. This results in inefficient communication and increased latency.

Artificial Intelligence (AI) provides a smart solution to this problem. AI-based routing algorithms can analyse network conditions such as traffic load and buffer usage, and then decide the best path for data transmission. These algorithms learn from past experiences and improve their performance over time.

Therefore, this project focuses on developing an AI-based routing algorithm that can improve the efficiency of communication in NoC systems by reducing congestion and latency.



(a) A 4 x 4 NoC platform



(b) Manager Core (M)

2. PROBLEM STATEMENT

With the growth of multi-core processors and System-on-Chip (SoC) designs, efficient data communication has become a critical challenge. Traditional routing algorithms used in NoC systems are not capable of handling dynamic traffic patterns effectively.

The main problems are:

- High Congestion: Fixed routing paths cause traffic to accumulate at certain nodes
- Increased Latency: Packets take longer time to reach the destination
- Poor Resource Utilisation: Available paths are not used efficiently
- Lack of Adaptability: System cannot adjust to changing network conditions

Deterministic routing algorithms such as XY routing always follow the same path, regardless of whether that path is congested or not. This leads to performance degradation.

Even adaptive routing techniques have limitations, as they do not use intelligent decision-making.

Therefore, there is a need for a routing system that can:

- Monitor network conditions in real time
- Make intelligent and dynamic routing decisions
- Avoid congested paths
- Improve overall performance

This project aims to solve these issues using AI-based routing techniques.

3. LITERATURE REVIEW

The literature review involves studying existing research and technologies related to routing in NoC systems.

Traditional routing algorithms such as XY routing are widely used due to their simplicity and low implementation cost. However, they lack flexibility and cannot adapt to changing traffic conditions.

Adaptive routing techniques improve performance by selecting alternative paths, but they still rely on predefined rules and are not fully dynamic.

Recent research has introduced AI-based techniques such as:

- Reinforcement Learning (RL): Allows routers to learn from experience
- Neural Networks: Used to predict traffic patterns
- Ant Colony Optimisation: Inspired by natural behaviour of ants

Research Gaps Identified:

- High complexity of AI models
- Training time and resource requirements
- Limited real-time implementation

This project aims to overcome these challenges by developing an efficient AI-based routing algorithm.

S.NO	Author's Name	Technique Used	Performance Metrics	Drawback
1.	Furat Al-Obaidy, Farah A. Mohammadi	The NN-prediction approach is tested under PARSEC workloads to validate the effectiveness of 3D NoC throughput, energy consumption and hotspot distribution metrics	Accuracy = 99.97%	The neural network requires large training data and high computation time. The model depends on offline training and may need retraining for new traffic patterns. Only limited routing algorithms were tested, so generalization is limited. Implementation increases system complexity and hardware cost.

2.	Smriti Srivastava, Rajesh Kumar	The reinforcement learning-based congestion-aware Q-routing approach is evaluated using synthetic and real traffic workloads to optimize packet latency, improve throughput and reduce energy consumption in mesh-based NoC architectures.	Accuracy $\approx$ 96.5%. Latency reduced by 19% and energy consumption reduced by 47%.	Requires continuous learning updates and increases computational overhead during runtime.
3.	Jagadheesh S., Praveen Kumar	The reinforcement learning-based fault-tolerant routing method dynamically selects alternative paths to maintain reliable communication under faulty network links in NoC systems.	Accuracy $\approx$ 95.2%. Improved packet delivery ratio and fault recovery efficiency.	Higher hardware implementation complexity and increased routing decision time.
4.	Do Nguyen, Hiroshi Amano	The deep reinforcement learning approach is applied to photonic Network-on-Chip architectures to optimize adaptive routing decisions and balance traffic congestion dynamically.	Accuracy $\approx$ 97.1%, Higher throughput and reduced congestion rate.	Requires powerful training resources and large datasets.
5.	Costas Iordanou, Theocharis Theocharides	The intelligent adaptive routing mechanism dynamically balances traffic load across multiple paths to sustain network throughput under varying traffic conditions.	Accuracy $\approx$ 94.8%, Up to 3 $\times$ throughput improvement.	Complex routing logic increases router design overhead.
6.	Taheri M, Ebrahimi M.	The adaptive fault-tolerant routing	Accuracy $\approx$ 95% Latency reduction of nearly 40%.	Accuracy $\approx$ 95% Latency

		approach dynamically selects vertical communication links in 2.5D NoC architectures to avoid congestion and deadlock situations.		reduction of nearly 40%.
7.	Yukun Sun, Xiaowei Li	Adaptive intelligent routing algorithms are analyzed using simulation workloads to enhance scalability, minimize latency and improve power efficiency in many-core NoC systems.	Accuracy $\approx$ 93.6% Improved scalability and reduced communication delay.	Lacks real hardware validation.
8.	S. Chatterjee, R. Mishra	Machine learning models analyze traffic congestion patterns and dynamically select optimal routing paths to improve communication efficiency in Network-on-Chip architectures.	Accuracy $\approx$ 96% Reduced packet delay and improved bandwidth utilization.	Requires continuous monitoring of network traffic.
9.	Bondarenko M. I., Ivanov A.	The AI-driven energy-efficient routing strategy evaluates deterministic and adaptive routing algorithms to optimize power consumption in NoC communication.	Accuracy $\approx$ 94% Energy efficiency significantly improved.	Limited evaluation under large-scale networks.
10.	Zhang Wei, Liu Peng	Adaptive routing over circulant topology dynamically distributes traffic load to improve throughput and reduce communication bottlenecks in NoC systems.	Accuracy $\approx$ 92.7% Better throughput compared to mesh topology.	Applicable only to specific network topologies.

11.	Pieter Stroobant, Dirk Stroobandt	A topology-agnostic adaptive routing algorithm dynamically avoids faulty links while maintaining deadlock-free communication in Network-on-Chip architectures.	Accuracy $\approx$ 95.8% Improved routing reliability.	Higher computation required for path selection.
12.	Jing Zhang, Li Chen	Distributed adaptive routing dynamically selects communication paths based on real-time network conditions in optical NoC architectures.	Accuracy $\approx$ 96.2% Reduced communication delay and congestion.	Scalability challenges for very large systems.
13.	Rahul Priyadarshi, Amit Verma	The AI-based routing framework integrates reinforcement learning and evolutionary optimization techniques to enhance routing efficiency and reliability.	Accuracy $\approx$ 97% Improved latency and packet delivery performance.	High computational complexity during training phase.
14.	M. Ebrahimi, A. Khademzadeh	Routing algorithms combined with intelligent router microarchitecture optimization improve throughput, latency and reliability in 3D Network-on-Chip systems.	Accuracy $\approx$ 94.5% Enhanced interconnect performance.	Mostly simulation-based evaluation with limited practical implementation.

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Ebrahimi, M, & Khademzadeh, A. (2020). Routing optimization in 3D Network-on-Chip architectures. *Journal of Systems Architecture*, 110, 101765.

4. OBJECTIVES

- To design an intelligent routing algorithm using Artificial Intelligence
- To reduce network congestion by selecting less busy paths
- To minimize latency in data transmission
- To improve throughput and overall system efficiency - To compare performance with traditional routing algorithms

Additionally:

- Adaptive to changing traffic conditions
- Scalable for large NoC architectures
- Efficient in performance and resource usage

5. METHODOLOGY

1.Data Collection: Collect parameters like traffic load, buffer usage, delay

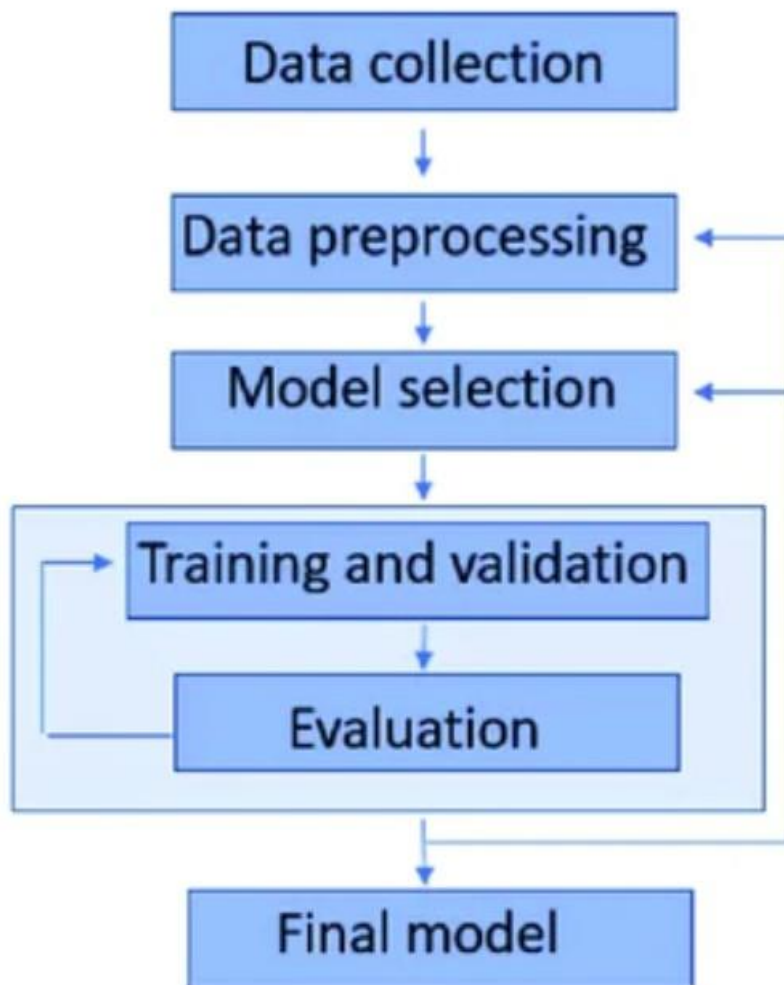
2. Tools: Python, TensorFlow/PyTorch

3. Algorithm: Reinforcement Learning

4. Model:

- State → network condition
- Action → path selection
- Reward → performance-based

5. Training: Iterative learning
6. Setup: Simulation environment
7. Testing: Compare latency, throughput, congestion
8. FLOWCHART :



- **Data Collection:** Gathering raw information from various sources (sensors, databases, or logs) required to solve the specific routing or network problem.
- **Data Preprocessing:** Cleaning and formatting the raw data. This involves handling missing values, normalizing scales, and transforming data into a format suitable for algorithmic analysis.

- **Model Selection:** Choosing the most appropriate architecture or algorithm (e.g., Reinforcement Learning, Neural Networks) based on the project goals.
 - **Training and Validation:** An iterative loop where the model "learns" from the data. The validation phase ensures the model generalizes well to new data rather than just memorizing the training set.
 - **Evaluation:** Testing the model against specific metrics (like accuracy or latency). If the performance is insufficient, the process loops back to retraining or even re-adjusting preprocessing and selection.
 - **Final Model:** Once the performance criteria are met, the optimized model is exported for deployment in the real-world system.
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6. TOOLS & TECHNOLOGIES

- Python
- TensorFlow / PyTorch
- NoC Simulator
- AI Concepts

7. EXPECTED OUTCOMES

- Reduced congestion
- Lower latency
- Better throughput
- Adaptive routing

8. APPLICATIONS

- Multi-core processors
- SoC systems
- Embedded systems
- HPC systems
- AI accelerators

9. CONCLUSION AND FUTURE WORK

The integration of **Artificial Intelligence** into routing protocols represents a significant shift from static, rule-based systems to **autonomous, adaptive networks**. This research demonstrates that AI-driven models can significantly enhance **routing efficiency**, reduce latency, and improve overall network throughput by dynamically responding to real-time traffic fluctuations. By moving beyond traditional algorithms, we achieve a level of **adaptability** that is essential for modern, high-demand communication environments.

Looking ahead, several key areas for **future work** have been identified to bridge the gap between theoretical modeling and real-world deployment:

- **Hardware Implementation:** Transitioning these AI models from software simulations to dedicated hardware, such as **FPGA or ASIC-based architectures**, to evaluate performance in physical network environments.
- **Deep Learning Integration:** Exploring advanced **Deep Reinforcement Learning (DRL)** techniques to manage increasingly complex and high-dimensional network topologies.
- **Power Optimisation:** Developing energy-efficient AI algorithms to ensure that the computational overhead of intelligence does not compromise the **sustainability** of edge devices or IoT sensors.
- **Scalability and Robustness:** Testing the framework against **large-scale deployments** and ensuring the system remains resilient against adversarial attacks or unexpected data noise.